

DATTA MEGHE COLLEGE OF ENGINEERING

(Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Airoli, Navi Mumbai | Website: www.dmce.ac.in

Ref No: DMCE/CIRCULAR/2026/01

Date: 04/10/2026

OFFICIAL CIRCULAR / NOTICE

MACHINE LEARNING (CS-401)

ISSUED BY: DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

AI COLLEGE OF ENGINEERING & TECHNOLOGY

Course: B.Tech Computer Science & Engineering
Subject: Machine Learning (CS-401)
Semester: V / VII
Assignment No: 1
Max Marks: 25
Target Audience: Students

Instructions:

1. Answer all questions clearly, providing diagrams or mathematical formulations where necessary.
2. Support your answers with relevant examples.
3. Submission Deadline: As communicated by the department portal.

Questions

Q1. Supervised vs. Unsupervised Learning (5 Marks)

What is the fundamental difference between supervised and unsupervised learning? Provide one real-world example for each to illustrate how they are applied in practice.

Q2. Bias-Variance Tradeoff (5 Marks)

Explain the bias-variance tradeoff. How does a model with high bias and high variance affect the overall performance of a machine learning algorithm?

Q3. Overfitting and Underfitting (5 Marks)

Define overfitting and underfitting in the context of model training. What specific techniques (such as regularization or cross-validation) can be used to prevent these issues?

Q4. Classification Metrics & Imbalanced Datasets (5 Marks)

Why is 'Accuracy' not always a reliable metric for classification problems, especially when dealing with highly imbalanced datasets? Define and explain the significance of Precision, Recall, and the F1-Score.

Q5. Gradient Descent Optimization (5 Marks)

Briefly explain how the Gradient Descent algorithm works. What is the role of the 'learning rate' in this optimization process, and what happens if it is set too high or too low?

Issued By:

Department of Computer Science & Engineering
AI College of Engineering & Technology

By Order,
Head of Department
Datta Meghe College of Engineering